

Renal Physiology

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GLOMERULAR STRUCTURE AND ULTRASTRUCTURE

Urine formation begins with the production of an ultrafiltrate of plasma. Chapter 1 details glomerular ultrastructure, and the essentials are revisited here. The ultrafiltration pathway consists of the fenestrated capillary endothelium, the capillary basement membrane (the noncellular layer), and the visceral epithelial cell layer (podocytes) of Bowman's capsule; the podocytes have large cell bodies and make contact with the basement membrane only by cytoplasmic foot processes. Parietal epithelial cells lining the inside of Bowman's capsule are highly proliferative and may provide a niche for replenishment of the podocyte pool, especially after injury.¹ Mesangial cells fill the intercapillary space and provide structural support for the glomerulus, including secretion of mesangial matrix. They are contractile and can affect glomerular capillary flow and the surface area for filtration.

Filtration is determined principally by the molecular size and shape of the filtered solute and to a lesser extent by its charge. The size cut-off is not absolute. Resistance to filtration begins at an effective molecular radius of approximately 2 nm, and substances with an effective radius greater than 4 nm are not filtered at all. The fenestrations between capillary endothelial cells have a diameter of 50 to 100 nm. The podocyte foot processes have gaps (filtration slits) with a diameter of 30 to 40 nm. These filtration slits are bridged by the slit diaphragms (SDs), which are themselves penetrated by small pores. The SDs likely constitute the main filtration barrier, although the endothelium (by preventing the passage of blood cells) and the basement membrane also contribute.² Furthermore, podocytes and endothelial cells are covered by a glycocalyx composed of negatively charged glycoproteins, glycosaminoglycans, and proteoglycans, and the basement membrane is rich in heparan sulfate proteoglycans. This accumulation of fixed negative charges restricts filtration of large negatively charged ions, mainly proteins (Fig. 2.1). Albumin is negatively charged and, with an effective radius of 3.6 nm (35 Å), is normally almost completely excluded from filtration. If the glomerular barrier loses fixed negative charge, as in some forms of early or mild glomerular disease (e.g., minimal change disease), albumin filterability increases and leads to proteinuria. It is possible that some albumin is normally filtered and is then almost completely reabsorbed along the proximal tubule, although this is controversial. Certainly proximal tubular cells can take up albumin via megalin and cubilin, transport proteins in the apical membrane.³ However, deletion of this tubular transport system in mice causes only a small increase in albuminuria, suggesting that filtration of albumin is both normally low and the major determinant of urinary excretion.⁴ Nonetheless, the filtration-to-reabsorption relationship is dynamic and physiologically regulated⁵ and may change in diseases such as diabetic nephropathy, contributing to albuminuria.⁶

The glomerular barrier is not a passive unidirectional filter: filtration pressure generates a small potential difference across the filtration

barrier, which may help clear the filter by driving negatively charged proteins out of the SD and back into the blood.⁷

GLOMERULAR FILTRATION RATE

The driving force for glomerular filtration (*net ultrafiltration pressure*) is determined by the sum of the hydrostatic and oncotic (colloid osmotic) pressure gradients between plasma and Bowman's space. Single-nephron glomerular filtration rate (SNGFR) is determined by the product of the net ultrafiltration pressure and the *ultrafiltration coefficient*, a composite of the surface area available for filtration (influenced by the contractile mesangium) and the hydraulic conductivity of the glomerular membranes, which is very high compared with other capillary beds. Thus, the SNGFR is calculated as:

$$K_f [(P_{gc} - P_{bs}) - (\pi_{gc} - \pi_{bs})]$$

where K_f is the ultrafiltration coefficient, P_{gc} is the glomerular capillary hydrostatic pressure (~45 mm Hg), P_{bs} is the Bowman space hydrostatic pressure (~10 mm Hg), π_{gc} is the glomerular capillary oncotic pressure (~25 mm Hg), and π_{bs} is Bowman's space oncotic pressure (0 mm Hg).

Net ultrafiltration pressure is approximately 10 mm Hg at the afferent end of the capillary tuft. As filtration of plasma from blood proceeds along the glomerular capillaries, proteins are concentrated and the glomerular capillary oncotic pressure (π_{gc}) increases. Loss of fluid would also be expected to cause a reduction in glomerular capillary hydrostatic pressure. Theoretically, toward the efferent end of a glomerular capillary, π_{gc} may equal the net hydrostatic pressure gradient, at which point ultrafiltration pressure would fall to zero. *Filtration equilibrium* is observed in a number of capillary beds (e.g., skeletal muscle). In the healthy glomerulus, however, *net ultrafiltration pressure* progressively declines and filtration equilibrium is never achieved for two reasons: first, the glomerular capillary bed flows into an arteriole, rather than a venous system, and efferent arteriole resistance limits the fall in capillary hydrostatic pressure (Fig. 2.2); second, progressive morphologic changes in the capillary endothelium reduce permeability toward the efferent end.

The total glomerular filtration rate (GFR) is the sum of the SNGFRs of functioning nephrons and the normal range is wide, typically cited at 120 mL/min per 1.73 m² surface area. GFR can be measured with renal clearance techniques. The renal clearance of any substance not metabolized by the kidneys is the volume of plasma required to provide that amount of the substance excreted in the urine per unit time. This is a virtual volume that can be expressed mathematically as follows:

$$C_y = U_y/P_y \times V$$

where C_y is the renal clearance of y ; U_y and P_y are the concentrations of y in the urine and plasma, respectively; and V is the urine flow rate. If a substance is freely filtered by the glomerulus and is not reabsorbed

Size and Charge Barrier

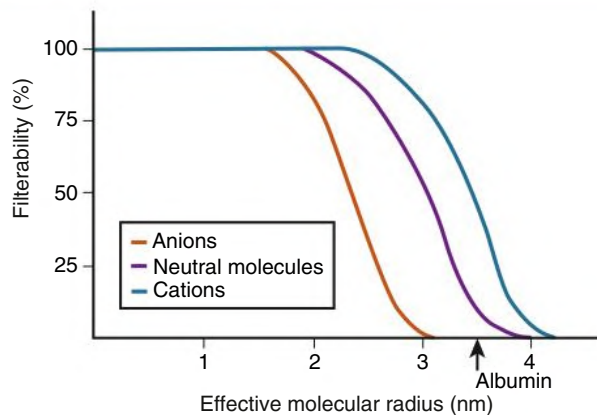


Fig. 2.1 Effects of Size and Electrical Charge on Filterability. Filterability of 100% indicates the substance is freely filtered; that is, its concentration in Bowman's space equals that in glomerular capillary plasma. For molecules and small ions (e.g., Na^+ , Cl^-), charge has no effect on filterability; but for ions whose effective molecular radius exceeds 1.6 nm, anions are filtered less easily than neutral molecules or cations. Albumin is an anion with a radius of 3.6 nm, and insignificant amounts of albumin are normally filtered. If the fixed negative charges of the glomerular basement membranes are lost, as in early minimal change nephropathy, charge no longer influences filterability; consequently, significant albumin filtration occurs.

Glomerular Filtration Pressures

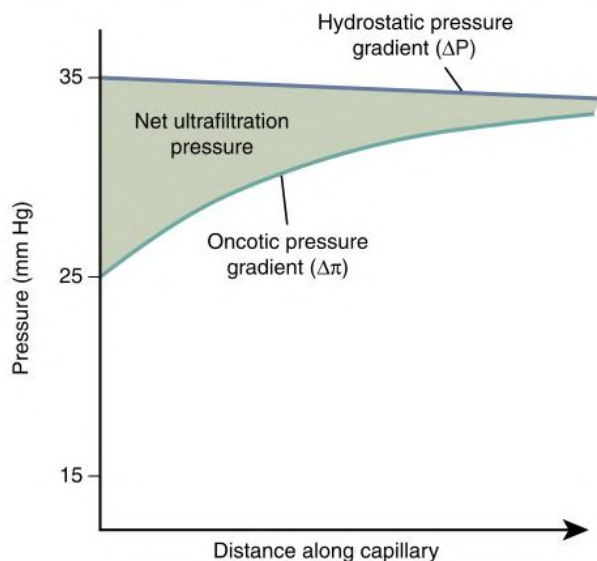


Fig. 2.2 Glomerular Filtration Pressures Along a Glomerular Capillary. Gating of flow by the efferent arteriole maintains a relatively constant hydrostatic pressure gradient ($\Delta P = P_{gc} - P_{bs}$) along the length of a capillary, whereas the opposing oncotic pressure gradient ($\Delta\pi = \pi_{gc}$) increases as protein-free fluid is filtered. The net ultrafiltration pressure falls along the capillary but does not fall to zero, the point of filtration equilibrium.

or secreted by the tubule, its renal clearance equals the GFR; that is, renal clearance measures the volume of plasma filtered through the glomeruli per unit time. The various methods for measuring GFR and their pitfalls are discussed in [Chapter 3](#).

RENAL PLASMA FLOW

The kidneys are approximately 0.5% of total body mass but account for approximately 7% of whole-body oxygen consumption. The renal blood flow (RBF) of approximately 1200 mL/min (~20% of the cardiac output) is delivered into the renal cortex, maintaining a high partial pressure of oxygen sufficient to support aerobic respiration by the proximal tubule. Almost all of the blood arriving by the renal artery is delivered into glomerular capillaries to support ultrafiltration; the *filtration fraction* (i.e., the proportion of plasma that is filtered—GFR/RPF) is 20% to 25%.

The glomerular capillaries drain into the efferent arteriole (rather than into the venous system), after which a second set of capillaries is formed. Capillaries arising from efferent arterioles of cortical (superficial and short-looped) nephrons form the peritubular network entwined around proximal and distal nephron segments in the cortex to provide an exit route for the bulk reabsorption of glomerular filtrate. The efferent arterioles of juxtamedullary (deeper and long-looped) nephrons provide descending vasa recta capillaries and supply blood to the medulla.

RBF can be calculated from measurement of renal plasma flow (RPF; typically ~650 mL/min) and hematocrit. RPF is measured by the clearance technique using substances that undergo both glomerular filtration and almost complete (or *effective*) tubular secretion. Para-aminohippuric acid (PAH, hippurate) is widely used for this purpose. PAH is an organic acid filtered by the glomerulus and then actively secreted into the urine through organic anion transporters (OATs) in the proximal straight tubule. This method is limited because the renal extraction of PAH is always less than 100% and variably underestimates RPF. This effect is compounded in patients with liver or kidney failure, where production of toxins and weak organic acids can interfere with PAH secretion, or during treatment with certain drugs, such as probenecid, which compete with PAH for tubular secretion and reduce PAH clearance. Thus, in a variety of clinical settings, PAH clearance is influenced by the abundance/pharmacokinetics of the OAT pathway and may not accurately reflect true RPF. Given that PAH clearance is also invasive (requiring intravenous [IV] infusion), time consuming, and expensive, alternative imaging approaches are being refined to estimate RBF directly, including Doppler ultrasound⁸ and magnetic resonance imaging.⁹

AUTOREGULATION OF RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

Acute physiologic variations in arterial blood pressure cause corresponding changes in RBF and GFR. These are short-lived because compensatory mechanisms return both RBF and GFR toward normal within seconds.¹⁰ In normal physiology, RBF and GFR are *autoregulated* and renal hemodynamics are largely independent of systemic blood pressure ([Fig. 2.3](#)). Autoregulation of RBF occurs throughout the preglomerular arterial system but is achieved primarily by the afferent arterioles. Autoregulation results from two main mechanisms: **Myogenic reflex**: Increased renal perfusion pressure stretches arteries and afferent arterioles, depolarizing smooth muscle cells to promote constriction of the vessel wall.

Tubuloglomerular feedback (TGF): Increased renal perfusion pressure increases sodium chloride (NaCl) delivery to the macula densa and specialized cells at the junction of the loop of Henle and distal nephron. The macula densa senses increased delivery and responds by causing afferent arteriolar constriction, reducing SNGFR.

These mechanisms are tonically active and dynamic and combine to restore both RBF and P_{gc} toward normal, reversing the initial

Renal Autoregulation

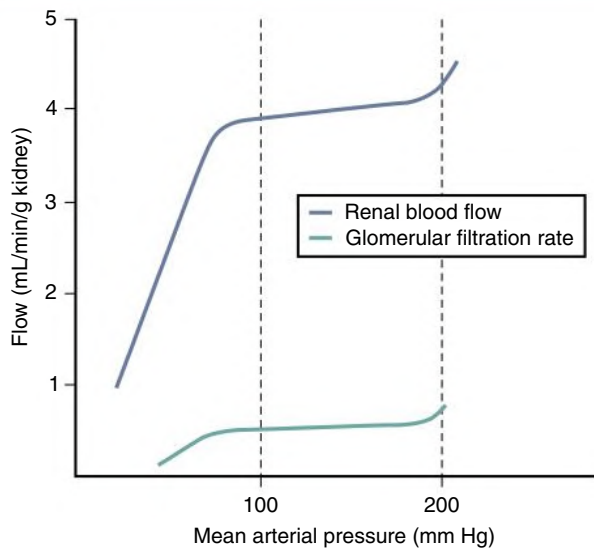


Fig. 2.3 Renal Autoregulation of Renal Blood Flow and Glomerular Filtration Rate. If mean arterial blood pressure is in the range of 80 to 180 mm Hg, fluctuations in blood pressure have only marginal effects on renal blood flow and glomerular filtration rate. This is an intrinsic mechanism and can be modulated or overridden by extrinsic factors. The dashed lines indicate the upper and lower limit of the autoregulatory range.

change in SNGFR. The TGF system is controlled within the juxta-glomerular apparatus (see [Chapter 1](#)), which consists of the macula densa region, the adjacent glomerulus, and afferent and efferent arterioles ([Fig. 2.4](#)). Both myogenic and TGF mechanisms are dependent on extracellular adenosine triphosphate (ATP) signaling. ATP is released from cells during vascular stretch, inducing vasoconstriction via P2 purinoceptors.¹¹ ATP is also the primary mediator of TGF. Increased NaCl delivery to the macula densa leads to increased NaCl uptake by these cells through a furosemide-sensitive $\text{Na}^+, \text{K}^+, 2\text{Cl}^-$ cotransporter, triggering ATP release into the surrounding extracellular space.¹² ATP can have a direct vasoconstrictor effect on the afferent arteriole, activating P2X₁ purinoceptors to depolarize the smooth muscle cells. However, because the macula densa is anatomically separated from the afferent arteriole by the extraglomerular mesangium, the final signal for TGF is likely to be adenosine, rather than ATP, acting on afferent arteriolar A₁ receptors to cause vasoconstriction.¹² The sensitivity of TGF is modulated by locally produced angiotensin II (Ang II), nitric oxide (NO), and certain eicosanoids (see later discussion).

The TGF regulation of GFR may be more complex than usually described, with evidence for regulatory cross-talk between the distal nephron and vasculature sites beyond the macula densa,¹³ as well as for synchronization of blood flow across networks of nephrons in response to changes in sodium delivery.¹⁴

Renal autoregulation is not perfect and renal hemodynamics are modulated by neural and humoral factors. Independent or unequal changes in the resistance of afferent and efferent glomerular arterioles, together with alterations in K_f (thought to result largely from mesangial cell contraction/relaxation), can result in disproportionate, or even contrasting, changes in RBF and GFR. In addition, changes in regional vascular resistance can alter the distribution of blood flow within the kidney. For example, contraction of pericytes in the descending vasa recta causes medullary vasoconstriction.¹⁵ Whole-kidney blood flow

Tubuloglomerular Feedback

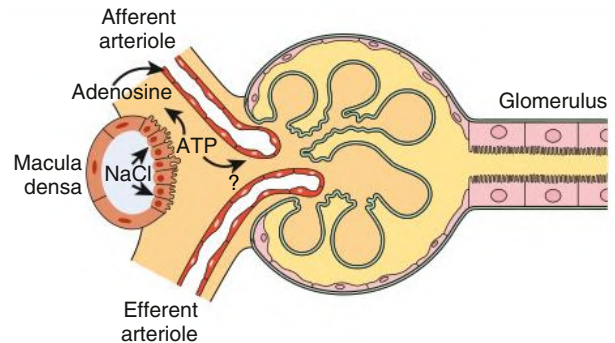


Fig. 2.4 Tubuloglomerular Feedback. Changes in the delivery of NaCl to the macula densa region of the thick ascending limb of the Henle loop cause changes in the afferent arteriolar caliber. The response is mediated by adenosine triphosphate (ATP), either directly or after metabolism to adenosine, and modulated by other locally produced agents such as angiotensin II and nitric oxide. Increased macula densa NaCl delivery results in afferent arteriolar constriction, thereby reducing the glomerular filtration rate (GFR).

Glomerular Hemodynamics

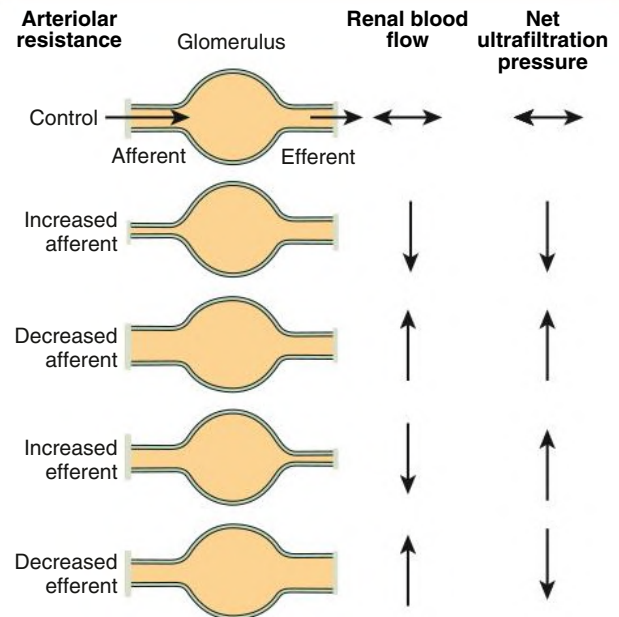


Fig. 2.5 Glomerular Hemodynamics. Changes in afferent or efferent arteriolar resistance will alter renal blood flow and (usually) net ultrafiltration pressure. The effect on ultrafiltration pressure, however, depends on the relative changes in afferent and efferent arteriolar resistance. The overall effect on the glomerular filtration rate will depend not only on renal blood flow and net ultrafiltration pressure but also on the ultrafiltration coefficient (K_f ; see [Table 2.1](#)).

may not change because blood can be diverted through the cortex. Sustained vasa recta constriction renders the medulla vulnerable to ischemic injury.¹⁶ [Fig. 2.5](#) indicates how changes in afferent and efferent arteriolar resistance can affect net ultrafiltration. [Table 2.1](#) lists some of the vasoactive factors that can alter renal hemodynamics (see [Integrated Control of Renal Function](#)). Damage to the afferent arteriole, as in patients with hypertension and progressive kidney disease, can impair renal autoregulatory mechanisms.

TABLE 2.1 Physiologic and Pharmacologic Influences on Glomerular Hemodynamics

	ARTERIOLAR RESISTANCE		Renal Blood Flow	Net Ultrafiltration Pressure	K_f	GFR
	Afferent	Efferent				
Renal sympathetic nerves	↑↑	↑	↓	↓	↓	↓
Epinephrine	↑	↑	↓	→	?	↓
Adenosine	↑	→	↓	↓	?	↓
Cyclosporine	↑	→	↓	↓	?	↓
NSAIDs	↑↑	↑	↓	↓	?	↓
Angiotensin II	↑	↑↑	↓	↑	↓	↓→
Endothelin-1	↑	↑↑	↓	↑	↓	↓
High-protein diet	↓	→	↑	↑	→	↑
Nitric oxide	↓	↓	↑	?	↑	↑ (?)
ANP (high dose)	↓	→	↑	↑	↑	↑
PGE_2/PGL_2	↓	↓ (?)	↑	↑	?	↑
Calcium channel blockers	↓	→	↑	↑	?	↑
ACE inhibitors, ARBs	↓	↓↓	↑	↓	↑	? ^a

The overall effect on GFR will depend on renal blood flow, net ultrafiltration pressure, and the ultrafiltration coefficient (K_f), which is controlled by mesangial cell contraction and relaxation. The effects shown are those seen when the agents are applied (or inhibited) in isolation; the actual changes that occur are dose dependent and are modulated by other agents.

^aIn clinical practice, GFR is usually either decreased or unaffected.

ACE, Angiotensin-converting enzyme; ARBs, angiotensin receptor blockers; ANP, atrial natriuretic peptide; GFR, glomerular filtration rate; NSAIDs, nonsteroidal anti-inflammatory drugs; PGE_2/PGL_2 , prostaglandins E_2 and I_2 .

TUBULAR TRANSPORT

Vectorial transport is the net movement of substances from tubular fluid to blood (reabsorption) or vice versa (secretion). The cell membrane facing the tubular fluid (*luminal* or *apical*) must have different properties from the membrane facing the blood (*peritubular* or *basolateral*). Such epithelia are “polarized,” allowing the net movement of substances across the cell (transcellular route). The *tight junction*, which is a contact point close to the apical side of adjacent cells, limits water and solute movement between cells (paracellular route).

Solute transport across cell membranes uses either passive or active mechanisms.

Passive Transport

Simple diffusion occurs down an electrochemical gradient, which is a composite of the concentration and electrical gradients (electrochemical gradient). With an undissociated molecule, only the concentration gradient is relevant; for a charged ion, the electrical gradient must also be considered. Simple diffusion does not require a direct energy source, although active transport is usually necessary to establish and maintain the electrochemical gradient.

Facilitated diffusion (coupled or carrier-mediated diffusion) depends on an interaction of the molecule or ion with a specific membrane carrier protein that facilitates its passage across the cell membrane's lipid bilayer. In almost all cases of carrier-mediated transport in the kidney, two or more ions or molecules share the carrier: one moiety moves down its electrochemical gradient and the other(s) is transported “uphill,” against the gradient.

Diffusion through a membrane channel (or pore) formed by specific integral membrane proteins is a form of facilitated diffusion because it allows charged, polar, and lipophobic molecules to pass through the membrane at a high rate.

Active Transport

Ion movement directly against an electrochemical gradient requires a source of energy and is known as active transport. In cells, this energy is derived from ATP production and its hydrolysis. The most important active cell transport mechanism is the sodium pump, which extrudes sodium ions (Na^+) from inside the cell in exchange for potassium ions (K^+) from outside the cell. In the kidney, this process is confined to the basolateral membrane. The Na pump derives energy from the enzymatic hydrolysis of ATP and is more correctly named Na^+,K^+ -ATPase. It exchanges $3Na^+$ for $2K^+$ and is electrogenic because it extrudes a net positive charge from the cell; Na^+,K^+ -ATPase is an example of a *primary* active transport mechanism. Other well-defined primary active transport processes in the kidney are the H^+ -ATPase, important in proton secretion in the distal nephron, and the Ca^{2+} -ATPase, partly responsible for calcium reabsorption.

Activity of the basolateral Na^+,K^+ -ATPase underpins all of the passive transport processes outlined earlier. It keeps intracellular Na^+ concentration low (10–20 mmol/L) and K^+ concentration high (~150 mmol/L), compared with their extracellular concentrations (~140 and 4 mmol/L, respectively). The *pump-leak* model of sodium transport uses the electrochemical gradient established and maintained by the Na pump to allow “leak” of Na^+ into the cell through a variety of membrane transport proteins, ranging from Na^+ channels to carrier proteins that couple Na^+ entry to the influx (*symport* or *cotransport*) or efflux (*antiport* or *countertransport*) of other molecules or ions. In various parts of the nephron, glucose, phosphate, amino acids, K^+ , and chloride ions (Cl^-) are all cotransported with Na^+ ; moreover, H^+ and Ca^{2+} can be countertransported against Na^+ entry. In each case, the non- Na molecule or ion is transported against its electrochemical gradient, using energy derived from the “downhill” movement of Na^+ . Their ultimate dependence on the Na^+,K^+ -ATPase makes them *secondary* active transporters.

Major Transport Mechanisms Along the Nephron

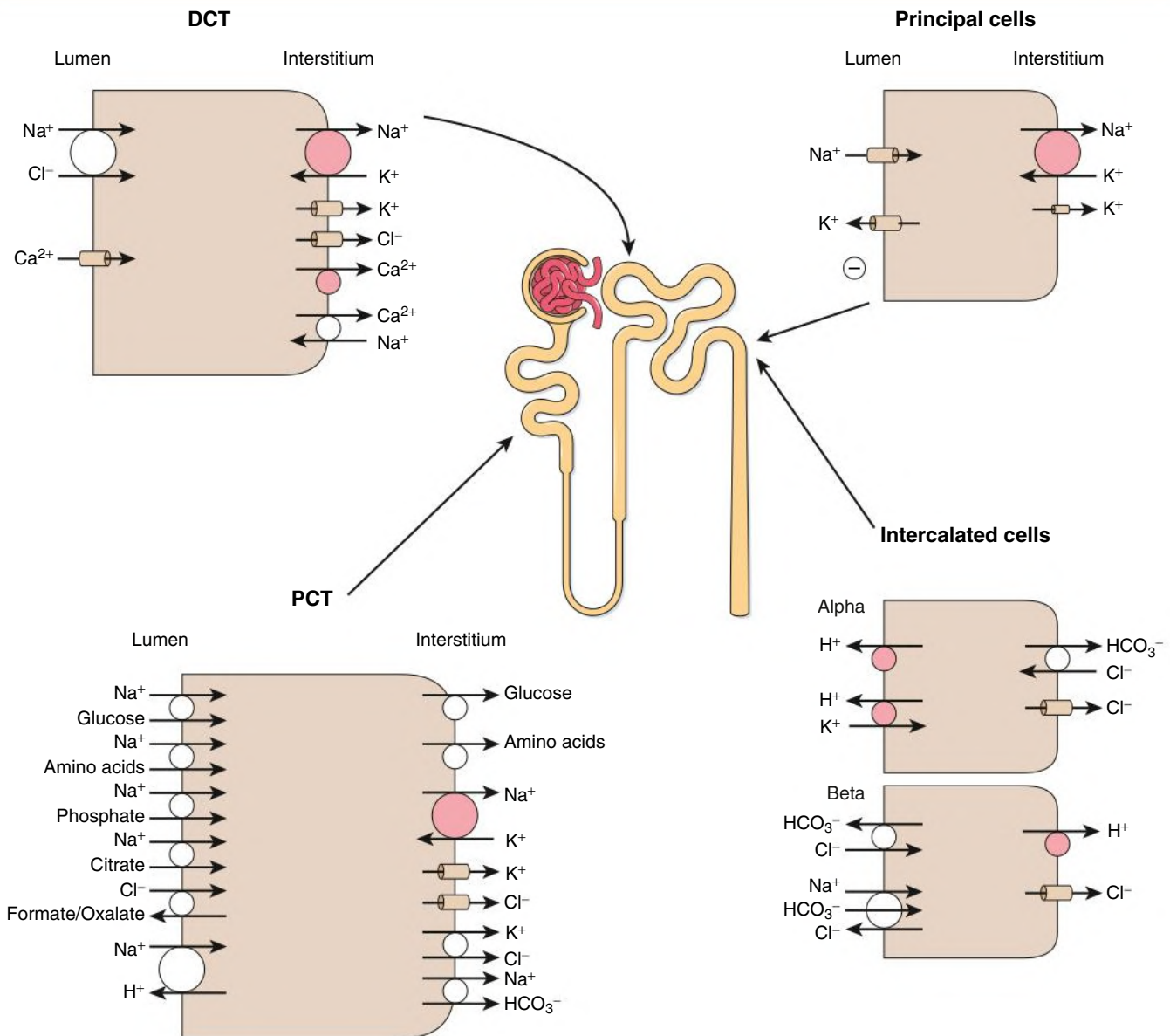


Fig. 2.6 Major Transport Mechanisms Along the Nephron. Major transport proteins for solutes in the apical and basolateral membranes of tubular cells in specific regions of the nephron. Stoichiometry is not indicated; it is not 1:1 in all cases. *Red circles* represent primary active transport; *white circles* represent carrier-mediated transport (secondary active); *cylinders* represent ion channels. In the proximal convoluted tubule (PCT), Na^+ enters the cell through an Na^+ - H^+ exchanger and a series of cotransporters. In the distal convoluted tubule (DCT), Na^+ enters the cell through the thiazide-sensitive Na^+ - Cl^- cotransporter. In the principal cells of the cortical collecting duct, Na^+ enters through the epithelial sodium channel (ENaC). In all cases, Na^+ is extruded from the cells through the basolateral Na^+ , K^+ -ATPase. Transporters in the thick ascending limb of Henle are dealt with separately (see Fig. 2.10).

TRANSPORT IN SPECIFIC NEPHRON SEGMENTS

In total, 180 L of plasma (largely protein free) is filtered each day, necessitating massive reabsorption by the nephron. Fig. 2.6 shows major transport mechanisms along the nephron; the loop of Henle is discussed separately.

Proximal Tubule

Proximal tubule cells are adapted for bulk reabsorption of the glomerular filtrate. The apical membrane have microvilli (brush border)

to provide a large absorptive area, and the basolateral membrane has folds, also increasing its surface area. Abundant mitochondria are concentrated at the basolateral membrane to supply the Na^+ , K^+ -ATPase with ATP. Proximal tubule transport is heavily reliant on oxidative phosphorylation (aerobic metabolism) and is susceptible to hypoxia and mitochondrial dysfunction. Drugs that are toxic to mitochondria (e.g., tenofovir) can cause Fanconi syndrome (see Chapter 50), and mitochondrial protection strategies may have potential to improve function in some forms of acute kidney injury.¹⁷

The proximal convoluted tubule (PCT, *pars convoluta*) makes up the first two-thirds of the proximal tubule, and the final third is the proximal straight tubule (*pars recta*). The proximal tubular epithelium can also be subdivided into three segments based on subtle structural and functional differences: S_1 is the initial short segment of the PCT; S_2 , the remainder of the PCT and the cortical segment of the *pars recta*; and S_3 , the medullary segment of the *pars recta*.

The NHE3 isoform of the $\text{Na}^+\text{-H}^+$ exchanger (antiporter) is the main route of Na^+ entry into proximal tubular cells. A battery of specialized transporters is also expressed in the apical membrane, coupling Na^+ entry to other solutes. The proximal tubule accounts for the bulk of Na^+ , K^+ , Cl^- , and bicarbonate (HCO_3^-) reabsorption and the almost complete reabsorption of amino acids and low-molecular-weight proteins (e.g., retinol binding protein, α - and β -microglobulin) that have passed the filtration barrier. The proximal tubule reabsorbs almost all the filtered glucose via the sodium-glucose cotransporters (the SGLTs). SGLT1, which is also expressed in the small intestine, localizes to the apical membrane of the S_3 proximal tubule, reabsorbing approximately 10% of the filtered glucose. Exclusive to the kidney, SGLT2 is expressed predominantly in the brush border of S_1 and S_2 and is the dominant pathway for glucose reabsorption. SGLT2 inhibitors (gliflozins) cause glycosuria and polyuria and lower blood glucose levels and, to a small extent, blood pressure in diabetics. In animal models of diabetes, gliflozins reduce glomerular hyperfiltration by inhibiting TGF β , reduce autophagy of renal tubule cells, and lower albuminuria. SGLT2 inhibitors may also preserve mitochondrial function, contributing to their renoprotection.¹⁸ SGLT2 inhibitors are an important new class of antidiabetic and nephroprotective drug used mainly to treat type 2 diabetes, and recent clinical trials have demonstrated a wider benefit in nondiabetic kidney disease and in heart failure.

Many other filtered solutes are also reabsorbed in the proximal tubule (e.g., ~60% of calcium, ~80% of phosphate, and 50% of urea). Constitutive expression of aquaporin 1 (AQP1) water channels in both membranes confers a large hydraulic permeability to cells. Furthermore, the protein junctional complexes (such as Claudin 1 and Zonula Occludens 1) connecting tubular cells are “leaky,” facilitating the reabsorption of large amounts of sodium, potassium, and water. The substantial hydraulic permeability requires only a very small osmotic driving force (<5 mOsm/kg/ H_2O), and the osmolality of the tubular fluid changes little along the proximal tubule, referred to as “isosmotic” reabsorption.

In the final section of the proximal tubule (late S_2 and S_3), there is efficient secretion of weak organic acids and bases, including most diuretics, PAH (see Renal Blood Flow), nonsteroidal anti-inflammatory drugs (NSAIDs), antiretroviral drugs, and some antibiotics. The molecular machinery for secretion is complex, mediated by 13 members of the SLC22 family of transport proteins.¹⁹

Loop of Henle

Anatomically, the loop of Henle includes the *pars recta* of the proximal tubule, the thin descending and ascending limbs (thin ascending limbs are present only in long-looped nephrons), the thick ascending limb, and the macula densa. In addition to its role in further reabsorption of solutes (Na^+ , Cl^- , K^+ , Ca^{2+} , Mg^{2+}), the loop of Henle is responsible for the kidney's ability to generate a concentrated or dilute urine, described in detail later. The thick limb of Henle also produces *uromodulin* (Tamm-Horsfall protein), normally the most abundant protein in urine. Mutations in the encoding gene (*UMOD*) cause rare autosomal dominant kidney diseases with medullary cyst formation, hyperuricemia, and progressive loss of renal function. Several polymorphisms have been associated with an increased risk for hypertension and chronic kidney disease. A recent mendelian randomization study to assess causality reported that increased urinary levels are

causally linked to CKD and increased blood pressure.^{19a} The role of uromodulin is not fully understood, but studies in mice suggest that it activates $\text{Na}^+\text{,K}^+\text{,2Cl}^-$ cotransport to promote sodium reabsorption in the thick limb of Henle. Uromodulin may also be a constitutive urinary inhibitor of calcium stone formation and can protect the kidney from ascending urinary tract infections.²⁰

Distal Nephron

The distal tubule involves three segments. The distal convoluted tubule (DCT) reabsorbs NaCl via a thiazide-sensitive NaCl cotransporter (NCC) in the apical membrane.²¹ The connecting tubule (CNT) is a functional intermediate between the DCT and the initial cortical collecting duct (CD; see Fig. 2.6), which is formed of *principal* cells (i.e., the main cell type) and *intercalated* cells (i.e., inserted between). Principal cells are responsible for Na^+ reabsorption, K^+ secretion,²² and regulated water reabsorption (see later discussion). Na^+ enters the principal cell through apical epithelial sodium channels (ENaC) and exits by the basolateral $\text{Na}^+\text{,K}^+\text{-ATPase}$. This process is electrogenic and establishes a lumen-negative transepithelial potential difference. K^+ enters the principal cell by the same basolateral $\text{Na}^+\text{,K}^+\text{-ATPase}$ and leaves by K^+ transport pathways in both membranes; however, the relative depolarization of the apical membrane (caused by Na^+ entry through ENaC) favors K^+ secretion into the lumen, the major route for which is through renal outer medullary potassium (ROMK) channels.

Intercalated (IC) cells contribute to acid-base homeostasis. Type A (or α) IC cells secrete H^+ into the tubular fluid via $\text{H}^+\text{-ATPase}$ and $\text{H}^+\text{,K}^+\text{-ATPase}$. Type B (or β) IC cells mediate HCO_3^- secretion into the final urine, in exchange for Cl^- . An exchanger called pendrin, encoded by *SLC26A4*, mediates this process. Type B IC cells also express a Na^+ -dependent chloride-bicarbonate exchanger (NDCBE; *SLC4A8*). Functionally coupled with pendrin, NDCBE achieves electroneutral sodium transport across the type B cell (see Fig. 2.6).²³

In the medullary CD there are increasingly fewer IC cells, whereas the principal-like cells are modified such that they reabsorb Na^+ but, lacking apical K^+ channels, do not secrete K^+ .

Figs. 2.7 and 2.8 show the sites of Na^+ and K^+ reabsorption and secretion along the nephron. Table 2.2 outlines the pathophysiologic consequences of known genetic defects in some of the major transporters in the nephron (see Chapter 49 for details).

GLOMERULOTUBULAR BALANCE

The proportion of filtered Na^+ that is excreted in the urine is so small (normally <1%) that even small changes in the filtered load would cause potentially major changes in the amount excreted in the final urine. For example, if GFR were to increase by 10%, and the rate of reabsorption remained unchanged, Na^+ excretion would increase more than 10-fold. An intrinsic feature of tubular function, however, is that the extent of Na^+ reabsorption in a given nephron segment is proportional to the Na^+ delivery to that segment. This process is called *glomerulotubular balance*. In perfect balance, both the reabsorption and the excretion of Na^+ would change in exactly the same proportion as the change in GFR, but glomerulotubular balance is usually less than perfect. Most studies have focused on the proximal tubule where glomerulotubular balance stabilizes delivery of Na^+ and fluid to the distal nephron, permitting efficient secretion of K^+ and H^+ . However, Na^+ reabsorption in the thick limb of Henle and distal tubule is also delivery-dependent. This partly explains why diuretics acting on the proximal tubule are less effective compared with those acting more distally: with distal-acting diuretics, there is less scope further downstream for compensatory Na^+ reabsorption. This also explains why combining two diuretics (acting at different nephron sites) can cause a more striking diuresis and natriuresis.

Renal Sodium Handling

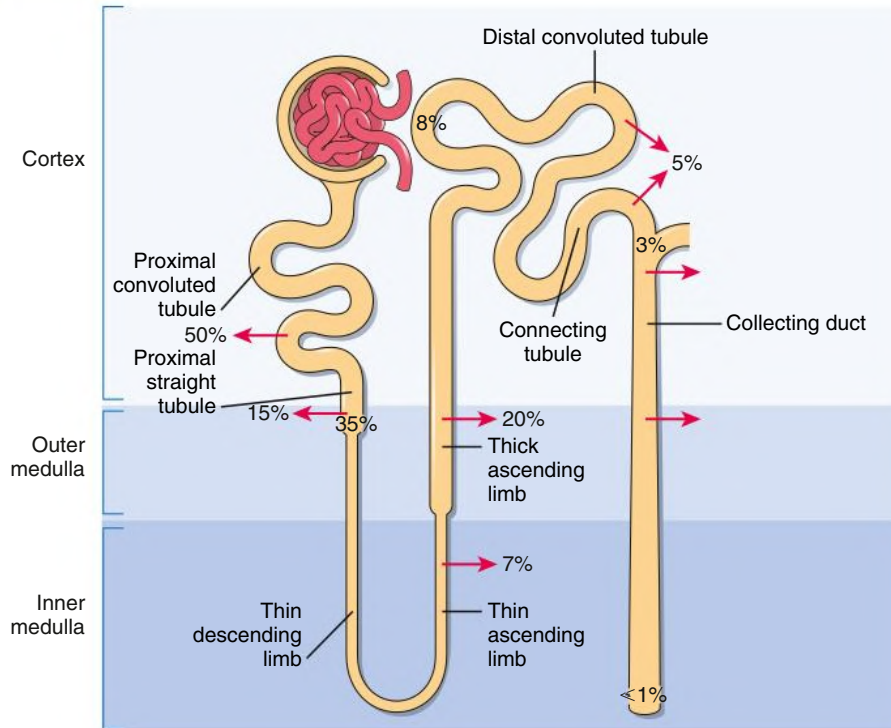


Fig. 2.7 Renal Sodium Handling Along the Nephron. Figures outside the nephron represent the approximate percentage of the filtered load reabsorbed in each region. Figures within the nephron represent the percentages remaining. Most filtered sodium is reabsorbed in the proximal tubule and loop of Henle; normal day-to-day control of sodium excretion is exerted in the distal nephron.

Renal Potassium Handling

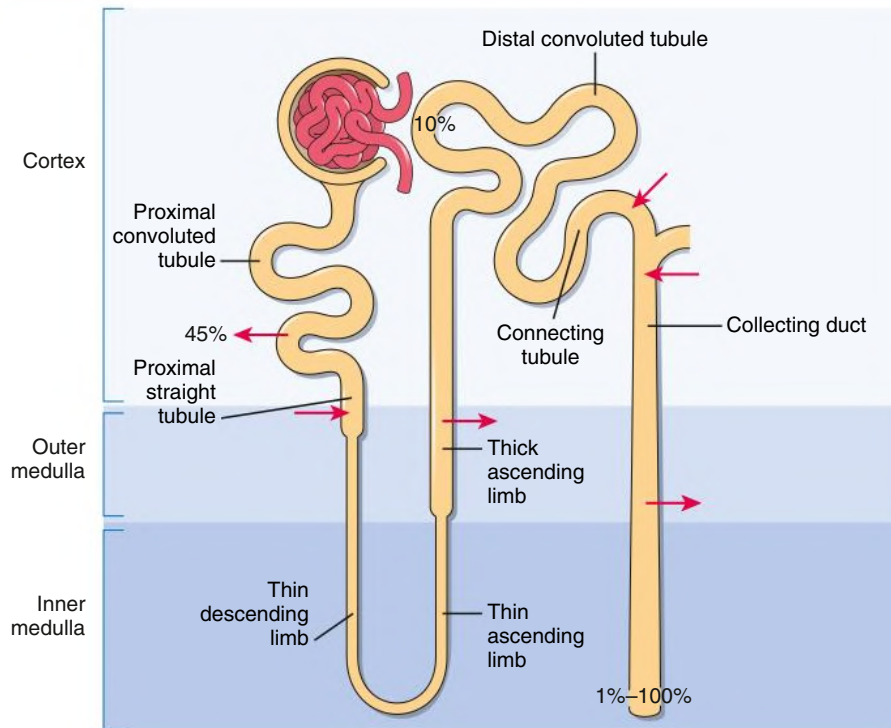


Fig. 2.8 Renal Potassium Handling Along the Nephron. Figures are not given for percentages reabsorbed or remaining in every region because quantitative information is incomplete, but most filtered potassium is reabsorbed in the proximal convoluted tubule and thick ascending limb of Henle; approximately 10% of the filtered load reaches the early distal tubule. Secretion by connecting tubule cells and principal cells in the late distal tubule–cortical collecting duct is variable and is the major determinant of potassium excretion.

TABLE 2.2 Genetic Defects in Transport Proteins Resulting in Renal Disease^a

Transporter	Consequence of Mutation
Proximal Tubule	
Apical Na ⁺ -cystine cotransporter	Cystinuria
Apical Na ⁺ -glucose cotransporter (SGLT2)	Renal glycosuria
Basolateral Na ⁺ -HCO ₃ ⁻ cotransporter	Proximal renal tubular acidosis
Intracellular H ⁺ -Cl ⁻ exchanger (ClC5)	Dent disease
Thick Ascending Limb	
Apical Na ⁺ -K ⁺ -2Cl ⁻ cotransporter	Bartter syndrome type 1
Apical K ⁺ channel	Bartter syndrome type 2
Basolateral Cl ⁻ channel	Bartter syndrome type 3
Basolateral Cl ⁻ channel accessory protein	Bartter syndrome type 4
Distal Convoluted Tubule	
Apical Na ⁺ -Cl ⁻ cotransporter	Gitelman syndrome
Collecting Duct	
Apical Na ⁺ channel (principal cells)	Overexpression: Liddle syndrome Underexpression: pseudohypoaldosteronism type 1b
Aquaporin 2 channel (principal cells)	Nephrogenic diabetes insipidus
Basolateral Cl ⁻ /HCO ₃ ⁻ exchanger (intercalated cells)	Distal renal tubular acidosis
Apical H ⁺ -ATPase (intercalated cells)	Distal renal tubular acidosis (with or without deafness)

^aFor more detailed coverage of these clinical conditions, see [Chapter 49](#).

ATP, Adenosine triphosphate.

The mechanism of glomerulotubular balance is not fully understood. In the proximal tubule, physical factors (Starling forces) operating across peritubular capillary walls may be involved. Glomerular filtration of an essentially protein-free fluid means that the plasma leaving the glomeruli in the efferent arterioles and supplying the peritubular capillaries has a relatively high oncotic pressure, favoring reabsorption of fluid from the proximal tubules. If GFR were reduced in the absence of a change in RPF, the *filtration fraction* would fall. Peritubular capillary oncotic pressure would also be reduced, and the tendency of the peritubular vasculature to take up fluid reabsorbed from the proximal tubule would be diminished. Backflux of this fluid is thought to occur through the leaky tight junctions, reducing net reabsorption ([Fig. 2.9](#)). This mechanism could only work, however, if GFR changed in the absence of a corresponding change in RPF; if the two changed in parallel, filtration fraction would stay constant, with no change in oncotic pressure.

A second contributory factor to glomerulotubular balance in the proximal tubule could be the filtered loads of glucose and amino acids: if more is filtered, the rates of Na⁺-coupled glucose and amino acid reabsorption in the proximal tubule will also increase. It has been proposed that the brush border microvilli of the proximal tubule serve a “mechanosensing” function, transmitting changes in torque (caused by altered tubular flow rates) to the tubular cells’ actin cytoskeleton

Peritubular Capillaries Modulate Fluid Reabsorption

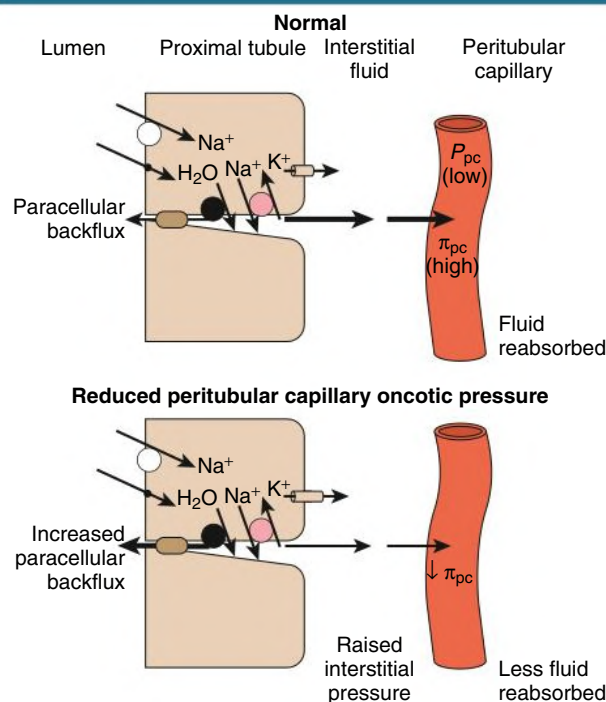


Fig. 2.9 Physical Factors and Proximal Tubular Reabsorption. Influence of peritubular capillary oncotic pressure on net reabsorption in proximal tubules. Uptake of reabsorbate into peritubular capillaries is determined by the balance of hydrostatic and oncotic pressures across the capillary wall. Compared with those in systemic capillaries, the peritubular capillary hydrostatic (P_{pc}) and oncotic (π_{pc}) pressures are low and high, respectively, so that uptake of proximal tubular reabsorbate into the capillaries is favored. If peritubular capillary oncotic pressure decreases (or hydrostatic pressure increases), less fluid is taken up, interstitial pressure increases, and more fluid may leak back into the lumen paracellularly; net reabsorption in proximal tubules would therefore be reduced.

and thereby modulating transporter activity.²⁴ The mechanisms are unknown, but the release of paracrine mediators such as ATP, dopamine, or Ang II into the lumen fluid may contribute.²⁵

Although renal sympathetic nerves and certain hormones can influence reabsorption in the proximal tubule and loop of Henle, under normal conditions, the combined effects of autoregulation and glomerulotubular balance ensure a relatively constant load of glomerular filtrate is delivered to the distal tubule, which exert normal day-to-day control of Na⁺ excretion. Evidence indicates important roles for the late DCT²¹ and the CNT,²⁶ in addition to the CD.²² Aldosterone, secreted from the adrenal cortex, activates mineralocorticoid receptors within CNT cells and in principal cells, leading to generation and activation by phosphorylation of the regulatory protein serum- and glucocorticoid-inducible kinase 1 (SGK1). SGK1 increases the density of apical ENaC (see [Fig. 2.6](#)) and enhances the basolateral Na⁺,K⁺-ATPase. Overall, Na⁺ transport across the principal cell stimulated and the maximal depolarization of the apical membrane drives K⁺ secretion into the tubular fluid. The electrophysiologic coupling of Na⁺ reabsorption to K⁺ secretion explains why aldosterone promotes antinatriuresis and kaliuresis. Under conditions of volume depletion, however, aldosterone promotes sodium retention *without* promoting urinary potassium excretion. The mechanism underlying this *aldosterone paradox* relates to the major stimulus for aldosterone production.

During hypovolemia, the renin-angiotensin system drives aldosterone synthesis. The combination of high aldosterone and Ang II engages the WNK4 kinase network and activates NCC in the DCT, promoting potassium-sparing sodium reabsorption, while also inhibiting ROMK channels to limit potassium secretion.²¹ Ang II also activates NDBCE and pendrin in the type B IC cell to promote potassium-sparing sodium reabsorption. Conversely, when hyperkalemia is the major stimulus for aldosterone release, then NCC is inhibited, pushing sodium reabsorption downstream where it is coupled to potassium secretion in the principal cell. Even modest increases in plasma potassium inhibit NCC activity, accounting for the natriuretic and blood pressure-lowering effects of high dietary potassium intake.²⁷

Mineralocorticoid receptors have equal affinity *in vitro* for aldosterone and cortisol. The circulating concentrations of cortisol vastly exceed those of aldosterone, and *in vivo* specificity for aldosterone along the distal nephron is conferred by the enzyme 11 β -hydroxysteroid dehydrogenase 2, which inactivates cortisol in the vicinity of the receptor.²⁸ Mutations in the encoding gene, or inhibition of the enzyme by derivatives of glycyrrhetic acid (found in licorice), cause hypokalemic hypertension because of excessive and unregulated stimulation of Na⁺ transport²⁹ and salt appetite³⁰ by cortisol (see also [Chapter 49](#)).

COUNTERCURRENT SYSTEM

The loop of Henle generates and maintains the interstitial osmotic gradient from the renal cortex (~290 mOsm/kg) to the tip of the medulla (~1200 mOsm/kg). As indicated in [Chapter 1](#), the loops of Henle of superficial nephrons turn at the junction between outer and inner medulla, whereas those of deep nephrons (long-looped nephrons) penetrate the inner medulla to varying degrees. The loops of Henle as a whole reabsorb approximately 40% of filtered Na⁺, mostly in the proximal straight tubule and the thick ascending limb, and approximately 25% of filtered water in the proximal straight tubule and the thin descending limbs of deep nephrons. Evidence suggests that the thin descending limb of superficial nephrons is relatively impermeable to water.³¹ Both the thin ascending limb (found only in deep nephrons) and the thick ascending limb are essentially impermeable to water. Both segments reabsorb Na⁺. In the thin ascending limb, this is passive. The thick ascending limb operates an active pump-leak system: the basolateral Na⁺,K⁺-ATPase maintains the electrochemical driving force for passive Na⁺ entry from the lumen through the Na⁺,K⁺,2Cl⁻ cotransporter (NKCC2) and, to a smaller extent, the NHE3 ([Fig. 2.10](#)). NKCC2 is directly inhibited by loop diuretics such as furosemide and bumetanide. Na⁺ exits the cell through the Na⁺,K⁺-ATPase, and Cl⁻ and K⁺ exit via basolateral ion channels, and a K⁺-Cl⁻ cotransporter. K⁺ can also recycle back across the apical membrane through ion channels and this sustains operation of NKCC2 because the availability of K⁺ is a limiting factor for the transporter (K⁺ concentration in tubular fluid being lower than Na⁺ and Cl⁻). Potassium recycling helps to generate a lumen-positive transepithelial potential difference in the thick ascending limb, which drives additional Na⁺ reabsorption through the paracellular pathway; for each Na⁺ reabsorbed by the transcellular route, another is reabsorbed paracellularly (see [Fig. 2.10](#)). Other cations (K⁺, Ca²⁺, Mg²⁺) are also reabsorbed by this route. The reabsorption of NaCl along the thick ascending limb in the absence of significant water reabsorption means that the tubular fluid leaving this segment is hypotonic; the thick ascending limb is also known as the *diluting segment* for this reason.

The reabsorption in the thick ascending limb of solute without water generates a “horizontal” osmotic gradient of about 200 mOsm/kg between the tubular fluid and interstitium. This separation is the *single osmotic effect*. The U-shaped arrangement of the loop of Henle,

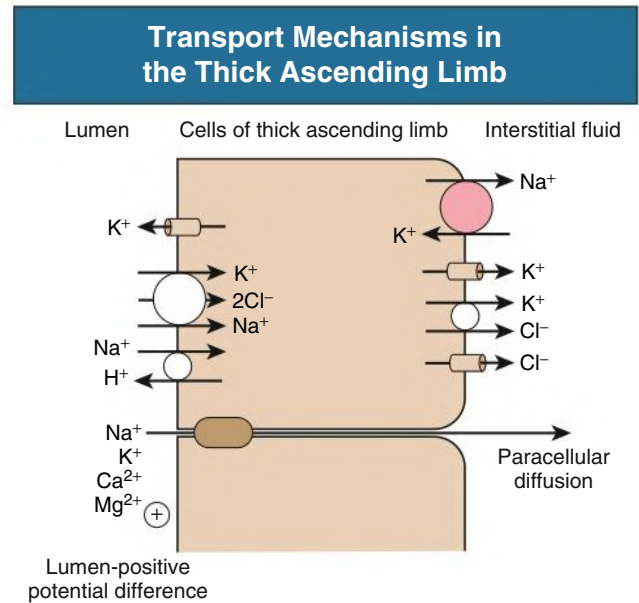


Fig. 2.10 Transport Mechanisms in the Thick Ascending Limb of Henle. The major cellular entry mechanism is the Na⁺-K⁺-2Cl⁻ cotransporter. The transepithelial potential difference drives paracellular transport of Na⁺, K⁺, Ca²⁺, and Mg²⁺.

in which flow in the descending limb is opposite to that in the ascending limb, multiplies the single effect to generate a much larger vertical (corticomedullary) osmotic gradient. This process is *countercurrent multiplication* ([Fig. 2.11](#)). Fluid entering the descending limb from the proximal tubule is isosmotic (~290 mOsm/kg). On encountering the hypertonicity of the medullary interstitial fluid (caused by NaCl reabsorption in the water-impermeable ascending limb), the fluid in the descending limb comes into rapid osmotic equilibrium, either by solute entry into the descending limb (superficial nephrons) or by water exit by osmosis (deep nephrons). These events, combined with continuing NaCl reabsorption in the ascending limb, result in a progressive increase in medullary osmolality from the corticomedullary junction to the papillary tip. A similar osmotic gradient exists in the thin descending limb, and at any level in the ascending limb, the osmolality of the tubule fluid is approximately 200 mOsm/kg less than in the surrounding tissue. The fluid that exits the loop of Henle has an osmolality of approximately 100 mOsm/kg. Further sodium is reabsorbed as the fluid traverses the distal nephron and CD, allowing for the formation of a very dilute urine (~50 mOsm/kg) if required for fluid balance.

Role of Urea

The thin limbs of the loop of Henle are relatively permeable to urea (ascending more permeable than descending), but the thick ascending limb and beyond are urea-impermeable up to the final section of the inner medullary CD. During antidiuresis, water reabsorption from the CDs concentrates urea, such that in the terminal inner medullary CD there is a large concentration gradient between the luminal fluid and interstitium. This section of the inner medullary CD expresses urea transporters (UT-A1 and UT-A3), allowing passive reabsorption of urea into the inner medullary interstitium. These processes are under the control of vasopressin (AVP, also known as antidiuretic hormone).³² The interstitial urea exchanges with vasa recta capillaries (see next section) and some urea enters the S₃ segment of the pars recta and the descending and ascending thin limbs; it is then returned to the inner medullary CDs to be reabsorbed. The net result of this urea recycling process is to add urea to the inner medullary interstitium, thereby

Countercurrent Multiplication

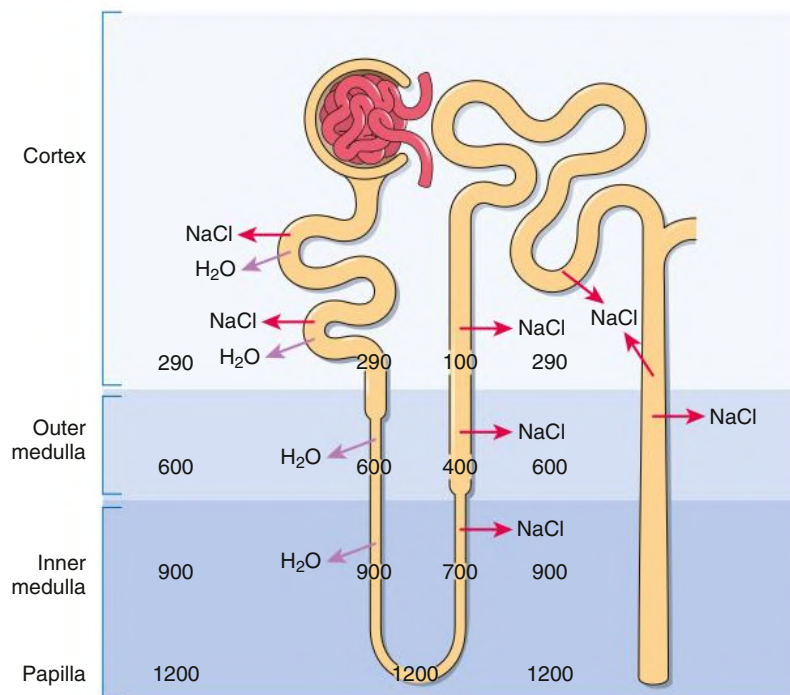


Fig. 2.11 Countercurrent Multiplication by the Loop of Henle. The nephron drawn represents a deep (long-looped) nephron. Figures represent approximate osmolarities (mOsm/kg). Osmotic equilibration occurs in the thin descending limb of Henle, whereas NaCl is reabsorbed in the water-impermeable ascending limb; hypotonic fluid is delivered to the distal tubule. In the absence of vasopressin, this fluid remains hypotonic during its passage through the distal tubule and collecting duct, despite the large osmotic gradient favoring water reabsorption. A large volume of dilute urine is therefore formed. During maximal vasopressin secretion, water is reabsorbed down the osmotic gradient, so that tubular fluid becomes isotonic in the cortical collecting duct and hypertonic in the medullary collecting duct. A small volume of concentrated urine is formed.

increasing interstitial osmolality. The high urea concentration within the medullary CD is balanced by a similarly high urea concentration in the medullary interstitium. This renders urea in the CD osmotically inactive and permits large quantities of urea to be excreted without osmotic diuresis. Moreover, the high urea concentration in the medullary interstitium should also increase osmotic water abstraction from the thin descending limbs of deep nephrons, thus raising the intraluminal Na^+ concentration within the thin descending limbs.

Although until recently this process was thought to drive passive Na^+ reabsorption along the thin ascending limbs, mice with genetic deletion of the urea transporters UT-A1 and UT-A3 have a greatly reduced urea concentration in the inner medullary interstitium but a normal interstitial NaCl gradient.³² Therefore, the mechanisms responsible for the inner medullary electrolyte gradients are still unclear. It is worth emphasizing, however, that the ultimate driving force for countercurrent multiplication is active Na^+ reabsorption in the thick ascending limb. For this reason, loop diuretics disrupt the osmotic gradient, and genetic mutations in the pathways contributing to efficient Na^+ reabsorption in the thick ascending limb cause salt-wasting Bartter syndrome (see Chapter 49).

Role of the Vasa Recta

The blood supply into the medulla arrives and leaves via the cortex. This U-shaped arrangement of the vasa recta prevents the dissipation of the medullary osmotic gradient by equilibration with the isotonic capillary blood. Solute enters and water leaves the vasa recta

as it descends into the medulla, and this is almost entirely offset by the opposite fluxes as the vasa recta ascends back to the cortex and drains into the arcuate vein. This passive process, called *countercurrent exchange* (Fig. 2.12), preserves medullary hypertonicity. Most solutes remain in the interstitium and any water reabsorbed from the collecting duct is swiftly removed, which results in a blood flow rate that is higher in the ascending than in the descending vasa recta.

Renal Medullary Hypoxia

Oxygen also undergoes countercurrent exchange by the medullary capillaries, diffusing from descending to ascending vasa recta and bypassing the deeper regions. This phenomenon, combined with ongoing energy-dependent Na^+ transport in the (outer medullary) thick ascending limb, renders medullary tissue relatively hypoxic. Thus, the partial pressure of oxygen normally decreases from about 50 mm Hg in the cortex to 10 mm Hg in the inner medulla.³³ Medullary cells are adapted to this relatively hypoxic environment, having a higher capacity for glycolysis than cortical cells. Moreover, a number of *heat shock proteins* are expressed in the medulla, which assist cell survival by restoring damaged proteins and by inhibiting apoptosis.³³

The degree of medullary hypoxia depends on the balance between medullary blood flow, influenced by contractile *pericytes* that surround the descending vasa recta, and oxygen consumption in the thick ascending limb (e.g., inhibition of NKCC2 by furosemide increases medullary oxygenation). In health, this balance is modulated by a variety of autocrine/paracrine agents (e.g., NO, eicosanoids, ATP, adenosine; see later

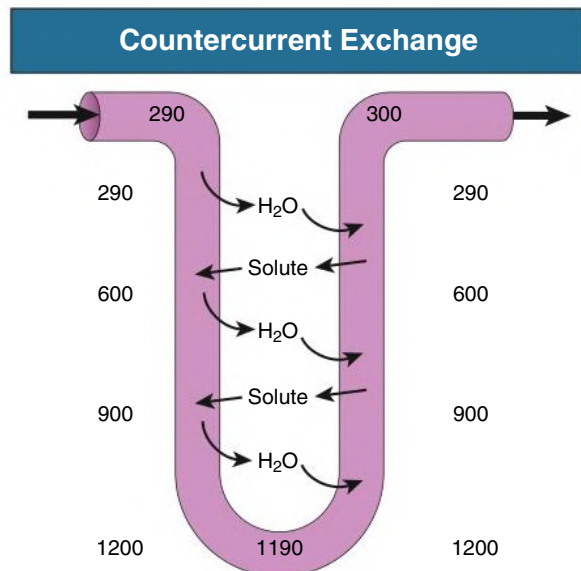


Fig. 2.12 Countercurrent Exchange by the Vasa Recta. Figures represent approximate osmolarities (mOsm/kg). The vasa recta capillary walls are highly permeable, but the U-shaped arrangement of the vessels minimizes the dissipation of the medullary osmotic gradient. Nevertheless, because equilibration across the capillary walls is not instantaneous, a certain amount of solute is removed from the interstitium.

discussion) and by renal nerves, which can alter medullary oxygenation by simultaneously regulating pericyte contraction and thick ascending limb Na^+ transport. Some cases of radiocontrast-induced nephropathy result from a disturbance in the balance between oxygen supply and demand, with consequent hypoxic medullary injury in which the normal cellular adaptations are overwhelmed, with subsequent apoptotic and necrotic cell death.

VASOPRESSIN (ANTIDIURETIC HORMONE) AND WATER REABSORPTION

Vasopressin is a nonapeptide synthesized in specialized neurons of the supraoptic and paraventricular nuclei. Vasopressin is transported from these nuclei to the posterior pituitary and released in response to increases in plasma osmolality and decreases in blood pressure. Osmoreceptors are found in the hypothalamus, and there is also input to this region from arterial baroreceptors and atrial stretch receptors. The actions of vasopressin are mediated by three receptor subtypes: V_{1a} , V_{1b} , and V_2 . The V_{1a} receptors are found in vascular smooth muscle and are coupled to the phosphoinositol pathway; they cause an increase in intracellular Ca^{2+} resulting in contraction. V_{1a} receptors also have been identified in the apical membrane of several nephron segments; activation by luminal vasopressin can influence Na^+ transport in these segments. V_{1b} receptors are found in the anterior pituitary, where vasopressin modulates adrenocorticotrophic hormone release. V_2 receptors are found in the basolateral membrane of principal cells in the late distal tubule and the whole length of the CD; they are coupled by a G_s protein to cyclic adenosine monophosphate generation, which ultimately leads to the insertion of aquaporin 2 (AQP2) water channels into the apical membrane of this otherwise water-impermeable segment (Fig. 2.13). In the X-linked form of nephrogenic diabetes insipidus, the most common inherited form, the V_2 receptor is defective.³⁴

Several aquaporins have been identified in the kidney.³⁵ AQP1 is found in apical and basolateral membranes of all proximal tubules and of thin descending limbs of long-looped nephrons, conferring

high basal water permeability to these segments. AQP3 is constitutively expressed in the basolateral membrane of CNT cells and cortical and outer medullary principal cells. AQP4 is constitutively expressed in the basolateral membrane of outer medullary principal cells and inner medullary CD cells. The variable water permeability of the late distal tubule and CD is determined by AQP2. Acute vasopressin release causes shuttling of AQP2 from intracellular vesicles to the apical membrane, whereas chronically raised vasopressin levels increase transcription and translation of the gene encoding AQP2. Apical insertion of AQP2 allows reabsorption of water, driven by the high interstitial osmolality. Vasopressin also contributes directly to the gradient by stimulating Na^+ reabsorption in the thick ascending limb and urea reabsorption through the UT-A1 and UT-A3 transporters in the inner medullary CD. In the (rare) autosomal recessive and (even rarer) autosomal dominant forms of nephrogenic diabetes insipidus, AQP2 is abnormal and/or fails to translocate to the apical membrane.³⁴

More frequently, defects in AQP2 shuttling contribute to the urine-concentrating defects associated with both hypokalemia and hypercalcemia. With chronic hypokalemia, AQP2 expression in the CD is reduced, possibly reflecting the generalized suppression of proteins central to urine concentration³⁶ and a reduction in the medullary osmotic gradient. With hypercalcemia, an elevated concentration of Ca^{2+} in tubular fluid activates the calcium-sensing receptor in the apical membrane of the principal cell, preventing insertion of AQP2. This is thought to be a protective mechanism to hinder kidney stone formation.³⁷ In addition, stimulation of the calcium receptor in the basolateral membrane of the thick ascending limb (a receptor similar to that in the parathyroid glands) reduces transcellular solute flux by inhibiting NKCC2 and ROMK channels and also by direct inhibition of paracellular permeability.³⁷ Overall, this reduces the medullary osmotic gradient for water reabsorption.

INTEGRATED CONTROL OF KIDNEY FUNCTION

The kidneys regulate blood volume by controlling the sodium content of the body and thus the *effective circulating volume*, a conceptual volume reflecting the degree of fullness of the vasculature. Chapter 8 details the regulation of effective circulating volume, and this chapter now introduces some of the key mediator systems. The list is not exhaustive, and it is important to recognize that these systems are not static and do not operate in isolation. Rather, kidney salt and water transport (and indeed blood pressure) has circadian rhythmicity³⁸ and is influenced by a dynamic web of interlocking control systems operating at whole-body and local, intrarenal levels.

Pressure Natriuresis

Acute increases in arterial blood pressure induce natriuresis. This is the *pressure natriuresis* phenomenon and is considered important for long-term stability of effective circulating volume.²⁵ Pressure natriuresis occurs because autoregulation of RBF is not perfect, and a rise in renal perfusion pressure increases flow in the medullary vasa recta. Kidney volume is constrained by the renal capsule: elevated flow in the medulla causes a proportionate rise in *renal interstitial hydrostatic pressure* (RIHP), which reduces net reabsorption in the proximal tubule by increasing paracellular backflux through the tight junctions of the tubular wall (see Fig. 2.9) and by inactivating apical membrane sodium transport proteins. Increased RIHP also stimulates the production of nitric oxide (NO) and ATP to inhibit sodium reabsorption in the thick limb of Henle and distal nephron.³⁹

Another renal action of NO results from the presence of inducible (type II) nitric oxide synthase (iNOS) in glomerular mesangial cells. Local NO production counteracts the mesangial contractile response

Action of Vasopressin

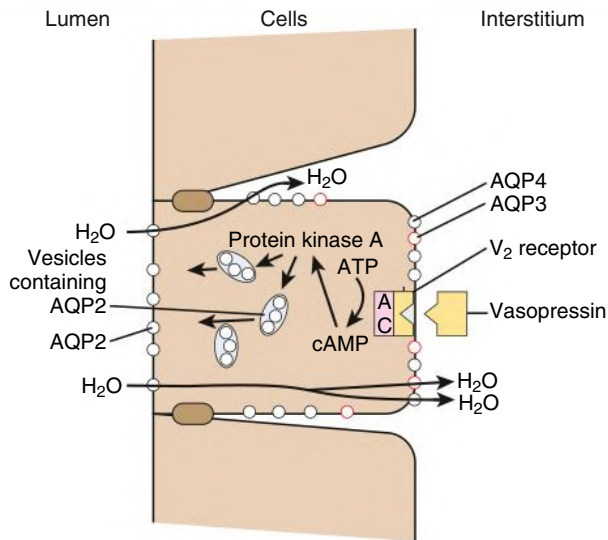


Fig. 2.13 Mechanism of Action of Vasopressin (Antidiuretic Hormone). Vasopressin binds to V_2 receptors on the basolateral membrane of collecting duct principal cells and increases intracellular cyclic adenosine monophosphate (cAMP) production, causing insertion of pre-formed aquaporin 2 (AQP2) water channels into the apical membrane through intermediate reactions involving protein kinase A. The water permeability of the basolateral membrane, which contains aquaporins 3 and 4, is permanently high. Therefore, vasopressin secretion allows transcellular movement of water from lumen to interstitium. AC, Adenylate cyclase; ATP, adenosine triphosphate.

to agonists such as Ang II and endothelin (see later discussion). Furthermore, NO may contribute to the regulation of medullary blood flow. Locally synthesized NO offsets the vasoconstrictor effects of other agents on the pericytes of the descending vasa recta, and it reduces Na^+ reabsorption in the thick ascending limb; both actions help protect the renal medulla from hypoxia.

Renal Sympathetic Nerves

Reductions in arterial pressure and/or central venous pressure suppress afferent signaling from arterial baroreceptors and atrial volume receptors, eliciting a reflex increase in renal sympathetic nervous discharge. This reduces urinary Na^+ excretion in at least three ways: (1) constriction of afferent and efferent glomerular arterioles (predominantly afferent), directly reducing RBF and GFR, and indirectly reducing RIHP; (2) direct stimulation of Na^+ reabsorption in the proximal tubule and the thick ascending limb of the Henle loop; and (3) stimulation of renin secretion by afferent arteriolar cells (see later discussion). Renal sympathetic overactivity has long been associated with Na^+ retention and experimental hypertension. The Symplicity-HTN trials initially indicated that bilateral sympathetic efferent denervation caused long-lasting reductions in blood pressure in patients with resistant hypertension. However, this early promise from small studies was not supported in a larger single-blind, sham-controlled trial, and enthusiasm for renal denervation to treat resistant hypertension waned. With advances in technology and refined study design, however, clinical trials such as SPYRAL-pivotal have shown reproducible and sustained reductions in ambulatory blood pressure, establishing confidence in the underlying concept. The next step is to identify those patients for whom renal denervation is clinically indicated and effective.

Renin-Angiotensin-Aldosterone System

The renin-angiotensin-aldosterone system (RAAS) is central to the control of extracellular fluid volume (ECFV) and blood pressure. Renin is synthesized and stored in specialized afferent arteriolar cells that form part of the juxtaglomerular apparatus and is released into the circulation in response to (1) increased renal sympathetic nervous discharge, (2) reduced stretch of the afferent arteriole after a reduction in renal perfusion pressure, and (3) reduced delivery of NaCl to the macula densa region of the nephron (see Fig. 2.4).

Renin catalyzes the production of the decapeptide Ang I from circulating angiotensinogen (synthesized in the liver). Ang I is converted by angiotensin-converting enzyme (ACE) to the octapeptide Ang II, which acts via AT1 receptors to regulate ECFV and blood pressure by:

- Inducing arteriolar vasoconstriction, including renal afferent and (particularly) efferent arterioles, thereby increasing arterial pressure but reducing RBF. The tendency of P_{gc} to increase is offset by Ang II-induced mesangial cell contraction, and reduced K_f and the overall effect on the GFR is unpredictable.
- Directly stimulating sodium reabsorption in the proximal tubule and distal tubule thiazide-sensitive NaCl cotransport.²⁷
- Stimulating aldosterone secretion from the zona glomerulosa of adrenal cortex. Aldosterone stimulates distal nephron sodium reabsorption.

Many tissues, including kidney, express a homologue of ACE called ACE2. This membrane-bound protein converts Ang II to Ang(1-7), thereby regulating local Ang II bioactivity. Ang(1-7) also activates the Mas receptor, causing vasodilation, natriuresis, and diuresis. ACE2 operates locally as a RAAS counterregulatory system and the balance between these two pathways determines the net effect on fluid-volume and blood pressure. ACE2 was also identified as a route for SARS-COV2 entry into cells.

Eicosanoids

Eicosanoids are a family of metabolites of arachidonic acid (AA) produced enzymatically by three systems: *cyclooxygenase*, with two isoforms, COX-1 and COX-2, both expressed in the kidney; *cytochrome P-450* (CYP-450); and *lipoxygenase*. The lipoxygenase system is activated (in leukocytes, mast cells, and macrophages) during inflammation and injury and is not considered further here.

The major renal eicosanoids produced by the COX system are the prostaglandins E_2 (PGE_2) and I_2 (PGI_2), both of which are renal vasodilators and buffer the effects of renal vasoconstrictor agents (e.g., Ang II, norepinephrine) and the vasoconstrictor thromboxane A_2 . PGE_2 and PGI_2 normally have minimal effects on renal hemodynamics, but during stressful situations such as hypovolemia, they help protect the kidney from excessive functional changes. Consequently, NSAIDs, which are COX inhibitors, can cause significant falls in GFR. PGE_2 also has tubular effects, inhibiting Na^+ reabsorption in the thick ascending limb of the Henle loop and both Na^+ and water reabsorption in the CD.⁴⁰ The action of PGE_2 in the thick ascending limb, together with a dilator effect on vasa recta pericytes, is another paracrine regulatory mechanism that helps protect the renal medulla from hypoxia. This may explain why inhibition of COX-2 can reduce medullary blood flow and cause apoptosis of medullary interstitial cells.

The metabolism of AA by renal CYP-450 enzymes yields epoxyeicosatrienoic acids (EETs), 20-hydroxyeicosatetraenoic acid (20-HETE), and dihydroxyeicosatrienoic acids (DHETs). These compounds appear to have multiple autocrine/paracrine/second messenger effects on the renal vasculature and tubules still to be fully unraveled. As with prostaglandins, EETs are vasodilator agents, whereas 20-HETE is a potent renal arteriolar constrictor and may be involved in the vasoconstrictor effect of Ang II and of the TGF mechanism. 20-HETE also constricts

vasa recta pericytes and may be involved in the control of medullary blood flow. Some evidence suggests that locally produced 20-HETE and EETs can inhibit sodium reabsorption in the proximal tubule and thick ascending limb. Indeed, CYP-450 metabolites of AA may contribute to the reduced proximal tubular reabsorption seen in pressure natriuresis.⁴¹

COX-2 is present in macula densa cells and has a critical role in the release of renin from juxtaglomerular cells (granular cells) in response to reduced NaCl delivery to the macula densa.⁴¹ A low-sodium diet increases COX-2 expression in the macula densa and simultaneously increases renin secretion; the renin response is virtually abolished in COX-2 knockout mice or during pharmacologic inhibition of COX-2. Therefore, it is likely that the low renin observed during administration of NSAIDs is largely a consequence of COX-2 inhibition. In addition to COX-2, the enzyme PGE synthase is expressed in macula densa cells, and the principal COX-2 product responsible for enhancing renin secretion is PGE₂, acting on specific receptors identified in juxtaglomerular cells. It is not clear whether PGI₂ is also synthesized in macula densa cells. nNOS (type I isomer) is also present in macula densa cells and produces NO that blunts TGF. NO also has a permissive role in renin secretion, although the mechanism is not understood. The increase in macula densa COX-2 expression induced by a low-sodium diet is attenuated during administration of selective nNOS inhibitors, which has led to speculation that NO is responsible for the increase in COX-2 activity and the resulting increase in juxtaglomerular renin secretion. Fig. 2.14 shows the established and proposed roles of COX-2 and nNOS in the macula densa.

Atrial Natriuretic Peptide

If blood volume increases significantly, the resulting atrial stretch stimulates the release of atrial natriuretic peptide (ANP) from atrial myocytes. This hormone increases sodium excretion by suppressing renin and aldosterone release and through a direct inhibitory effect on sodium reabsorption in the medullary CD. ANP may also increase GFR because high doses cause afferent arteriolar vasodilation and mesangial cell relaxation (thus increasing K_f ; see Table 2.1).

Endothelins

The kidney is a rich source of endothelins, predominantly paracrine peptides acting via ET_A and ET_B receptors: endothelin-1 (ET-1) is the major isoform.⁴² ETA activation causes sustained vasoconstriction and mesangial contraction, meaning that ET-1 can significantly reduce RBF and GFR (see Table 2.1). ET_A also reduces medullary blood flow and polarizes resident immune cells to a proinflammatory phenotype. In combination, these effects of ET_A activation explain why chronically elevated ET-1 may contribute to kidney diseases. Activation of ET_B inhibits ENaC-mediated sodium reabsorption, causing natriuresis, and mice with CD-specific knockout of ET_B receptors exhibit salt-sensitive hypertension. There is mounting evidence that NO mediates the natriuretic and diuretic effects of medullary ET_B stimulation,⁴² highlighting the potential importance of ET-1/NO interactions in the control of Na⁺ and water excretion. ET_A antagonists reduce albumin excretion in diabetic kidney disease⁴³ and are being investigated in other forms of kidney disease with proteinuria. Selective ET_A antagonists were thought to avoid the sodium/water retention and edema seen in trials with nonselective endothelin antagonists; however, they still seem to show this effect, especially at higher doses, suggesting that both ET_B and ET_A play a role in collecting duct sodium excretion.⁴⁴

ATP and Other Metabolic Intermediates

Paracrine signaling is firmly established as a key aspect of renal regulation. Multiple systems form an interlocking network of control that

Interactions Between Macula Densa and Afferent Arteriole

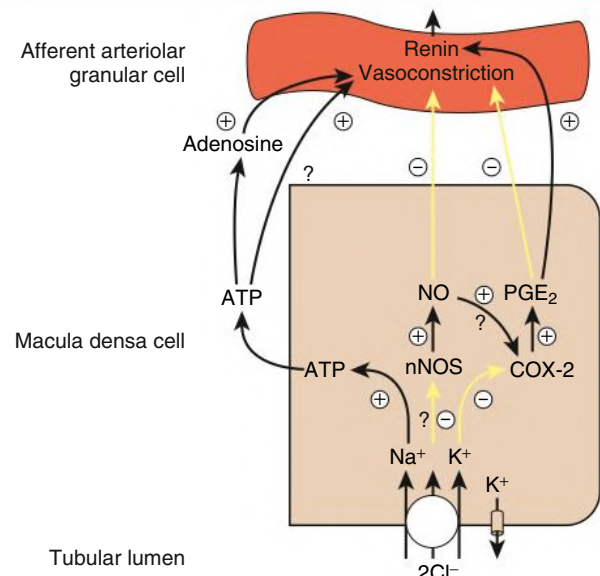


Fig. 2.14 Interactions Between Macula Densa and Afferent Arteriole: Proposed Mediators of Renin Secretion and Tubuloglomerular Feedback. Both cyclooxygenase-2 (COX-2) and neuronal nitric oxide synthase (nNOS) enzyme systems are present in macula densa cells. Increased NaCl delivery to the macula densa stimulates NaCl entry into the cells through the Na⁺-K⁺-2Cl⁻ cotransporter. This causes afferent arteriolar constriction through adenosine or adenosine triphosphate (ATP) and also inhibits COX-2 activity; the latter effect might be mediated partly through inhibition of (nNOS-mediated) nitric oxide (NO) production. Generation of prostaglandin E₂ (PGE₂) by COX-2 stimulates renin release. PGE₂ also modulates vasoconstriction, as does NO.

can rapidly adjust hemodynamic and tubular function; such systems support and interact with classical endocrine and neural control mechanisms. One of the most powerful paracrine regulators are the extracellular purines, ATP, adenosine diphosphate (ADP) and adenosine that alter cell function via specific cell surface receptors.¹² Purinoceptors are subdivided into P1 and P2 receptors. P1 receptors are responsive to adenosine and are more usually known as *adenosine receptors* (A₁, A_{2a}, A_{2b}, and A₃). P2 receptors are responsive to nucleotides (e.g., ATP, ADP) and are further subdivided into P2X (ligand-gated ion channel) and P2Y (G-protein coupled) receptors, each category having a number of subtypes.⁴⁵ As indicated earlier, A₁ and P2X₁ receptors are found in afferent arterioles and mediate vasoconstriction. Purinoceptors are also found in the apical and basolateral membranes of renal tubular cells. Stimulation of A₁ receptors enhances proximal tubular reabsorption and inhibits CD Na⁺ reabsorption, whereas stimulation of P2 receptors generally has an inhibitory effect on tubular transport.⁴⁵ Thus, luminally applied nucleotides, acting on a variety of P2 receptor subtypes, can inhibit Na⁺ reabsorption in the proximal tubule, distal tubule, and CD, and stimulation of P2Y₂ receptors in the CD inhibits vasopressin-sensitive water reabsorption.

Other metabolic intermediates regulate kidney function, and our understanding of their roles often relies on assigning a cognate ligand to an “orphan” G-protein coupled receptor. The Krebs cycle intermediate succinate stimulates renin release via GPR91, which may contribute to RAAS activation in type 2 diabetes.⁴⁶ Lactate, which accumulates in the kidney after ischemic injury, is the ligand for GPR81, which is

expressed in the renal vasculature. Activation of GPR81 stimulates ET-1 release, causing renal vasoconstriction.⁴⁷

Extracellular Vesicles and other Luminal Factors

All kidney cells release vesicles (including exosomes) and contain proteins, mRNA, and microRNA specific to their cell of origin and have generated much interest as a reservoir for kidney disease biomarker discovery. Vesicles released from one renal cell can be carried in the urine and taken up under hormonal control by downstream cells,

influencing their function.⁴⁸ Vesicles from distant organs, such as the liver and heart, are also able to transfer microRNA to kidney cells in vivo, but the full implications of this novel organ-to-organ communication system for health and disease are still unknown.⁴⁹ Finally, recent studies have also shown the importance and potential clinical relevance of enzymes (serine proteases) in urine that can activate ENaC and thereby affect Na^+ reabsorption, particularly in proteinuric diseases such as the nephrotic syndrome when plasma proteases can be filtered and enter the glomerular filtrate.⁵⁰

SELF-ASSESSMENT QUESTIONS

1. Filtration at the glomerulus is a passive process. Which of the following statements are *true*?
 - A. Albumin is too large to be freely filtered.
 - B. Albumin is not filtered because the glomerular barrier has a net positive charge.
 - C. Hydrostatic pressure in the glomerular capillaries is the only force influencing filtration rate.
 - D. Dilation of the afferent arteriole, but not the efferent arteriole, can increase filtration rate.
 - E. Mesangial cells do not influence filtration rate.
2. Fanconi syndrome is a disorder of the proximal tubule. Patients can present with glycosuria, polyuria, hypercalciuria, hyperphosphaturia, and low-molecular-weight proteinuria. These features can be explained by which of the following actions of the proximal tubule?
 - A. It reabsorbs about 100% of the filtered glucose.
 - B. It reabsorbs approximately 65% of the filtered water because of a large transepithelial osmotic gradient.
 - C. It actively reabsorbs calcium through Ca^{2+} -ATPase in the apical membrane.
 - D. It secretes phosphate through the organic anion transporter system.
 - E. It is a tight epithelium, impermeable to paracellular movement of proteins.
3. The corticomedullary osmotic gradient is required for urine concentration. Which of the following would diminish this gradient?
 - A. Increased circulating vasopressin.
 - B. Increased blood flow through the renal medulla.
 - C. Activation of the sympathetic nervous system.
 - D. Activation of the renin-angiotensin-aldosterone system.
 - E. Increased urea recycling through the collecting duct system.

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